Ambry Genetics
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FINAL REPORT - 10/31/2025

Ordered By	Contact ID:6464238	Org ID:5469	Patient Legal Name: **Ul Hasan, Sabah**
Medical Professional:	Munoz, Andrea, Pac		Accession #: **25-691252** Specimen #:
Client:	Sharp Healthcare - Health Center Dr. (08085)		AP2 Order #: 3161483 Specimen: Blood EDTA (Purple top)
Additional Authorized Recipient:			Birthdate: 11/27/1990 Sex assigned at birth: F
Burnett, Brittany MS, CGC			MRN #: 108053704 Collected: 10/13/2025
			Indication: Diagnostic/Family History Received: 10/14/2025
			Test Started: 10/14/2025

BRCA1/2 Analyses with CancerNext-Expanded® +RNAinsight®: Analyses of Genes Associated with Hereditary Cancer (77 genes)

RESULTS

SDHB	**Variant, Likely Pathogenic:**	**p.C68F**
TP53	Variant, Unknown Significance:	p.G154S

SUMMARY

POSITIVE: Likely Pathogenic Variant Detected

INTERPRETATION

- This individual is heterozygous for the **p.C68F (c.203G>T)** likely pathogenic variant in the *SDHB* gene.
- This result is consistent with a diagnosis of *SDHB*-related hereditary pheochromocytoma-paraganglioma.
- **Risk estimate**: increased lifetime risks for pheochromocytoma and paraganglioma (22-47%) and renal cell carcinoma (4-14%).
- The expression and severity of disease for this individual cannot be predicted.
- Genetic testing for likely pathogenic variants (VLPs) in family members can be helpful in identifying at-risk individuals.
- Genetic counseling is a recommended option for all individuals undergoing genetic testing.

This individual is also heterozygous for the p.G154S (c.460G>A) variant of unknown significance in the *TP53* gene, which may or may not contribute to this individual's clinical history. Refer to the supplementary pages for additional information on this variant. No additional pathogenic mutations, variants of unknown significance, or gross deletions or duplications were detected. Genes Analyzed (77 total): ***AIP, ALK, APC, ATM, BAP1, BARD1, BMPR1A, BRCA1, BRCA2, BRIP1, CDC73, CDH1, CDK4, CDKN1B, CDKN2A, CEBPA, CHEK2, DICER1, ETV6, FH, FLCN, GATA2, LZTR1, MAX, MEN1, MET, MLH1, MSH2, MSH6, MUTYH, NF1, NF2, NTHL1, PALB2, PHOX2B, PMS2, POT1, PRKAR1A, PTCH1, PTEN, RAD51C, RAD51D, RB1, RET, RPS20, RUNX1, SDHA, SDHAF2, SDHB, SDHC, SDHD, SMAD4, SMARCA4, SMARCB1, SMARCE1, STK11, SUFU, TMEM127, TP53, TSC1, TSC2, VHL*** **and** ***WT1*** **(sequencing and deletion/duplication);** ***AXIN2, CTNNA1, DDX41, EGFR, HOXB13, KIT, MBD4, MITF, MSH3, PDGFRA, POLD1*** **and** ***POLE*** **(sequencing only);** ***EPCAM*** **and** ***GREM1*** **(deletion/duplication only)**. RNA data is routinely analyzed for use in variant interpretation for all genes.

SDHB **Additional Information**

The **p.C68F** variant (also known as c.203G>T), located in coding exon 3 of the *SDHB* gene, results from a G to T substitution at nucleotide position 203. The cysteine at codon 68 is replaced by phenylalanine, an amino acid with highly dissimilar properties. Other variant(s) at the same codon, p.C68Y (c.203G>A), have been identified in individual(s) with features consistent with SDHB-related hereditary pheochromocytoma-paraganglioma (Burnichon N et al. *Hum. Mol. Genet.* 2011 Oct;20:3974-85; Loriot et al. *J Clin Endocrinol Metab.* 2012 Jun;97(6):E954-62; Bennedbæk M et al. *Hered Cancer Clin Pract.* 2016 Jun;14:13; Ben Aim L et al. *J Med Genet*, 2019 Aug;56:513-520; Main AM et al. *Endocr Connect*, 2020 Aug;9:793-803; Ambry internal data). This amino acid position is highly conserved in available vertebrate species. In addition, this alteration is predicted to be deleterious by *in silico* analysis. This variant is considered to be rare based on population cohorts in the Genome Aggregation Database (gnomAD). Based on the majority of available evidence to date, this variant is likely to be pathogenic.

The *SDHB* gene (NM_003000.2) is located on chromosome 1p36.13, encodes the succinate dehydrogenase complex iron sulfur subunit B, and contains 8 coding exons. Pathogenic variants in this gene are known to cause *SDHB*-related hereditary pheochromocytoma-paraganglioma (PCC-PGL), which is inherited in an autosomal dominant fashion, and *SDHB*-related mitochondrial complex II deficiency, which is inherited in an autosomal recessive fashion. Pathogenic variants in *SDHB* confer a significantly increased cumulative lifetime risk for paraganglioma or pheochromocytoma (22-47% lifetime risk), renal cell carcinoma (4-14% lifetime risk), gastrointestinal stromal tumors, and may also confer an increased risk for pituitary adenomas. *SDHB*-associated PGLs are characterized by a higher risk of aggressive behavior/malignancy and development of metastatic disease (Andrews KA et al. *J. Med. Genet*. 2018 06;55:384-394; Jochmanova I et al. *J. Cancer Res. Clin. Oncol*. 2017 Aug;143:1421-1435; Evenepoel L et al. *Genet Med*. 2015 Aug;17(8):610-20; Benn DE, et al. (2018) *J Med Genet* 55(11):729-734; White G, et al. (2022) *Endocr Connect* 11(2)). Pathogenic variants in this gene account for up to 25% of hereditary PCC-PGL syndrome (Andrews KA et al. *J Med Genet.* 2018 Jun;55(6):384-394). Penetrance in *SDHB*-related hereditary PCC-PGL is incomplete and variable expressivity is observed; therefore, cancer risks will differ based on individual and family history. Loss of function has been reported as the mechanism of disease for *SDHB*-related hereditary PCC-PGL. *SDHB*-related mitochondrial complex II deficiency is a highly variable inborn error of metabolism characterized by developmental delay, cardiomyopathy (with or without cardiac conduction defects), spasticity, abnormal brain imaging (such as leukoencephalopathy), or developmental regression following acute illness or infection. Additional features may include hypotonia, skeletal myopathy, or ophthalmological abnormalities (ptosis, ophthalmoplegia, pigmentary retinopathy, nystagmus, or optic atrophy). Onset is variable, and can occur in infancy with severe multisystem involvement or in adulthood with isolated cardiac or muscle involvement (Alston CL et al. *J Med Genet*. 2012 Sep;49(9):569-77; Fullerton M, et al. (2020) *Mol Genet Metab* 131(1):53-65). Biallelic loss of function has been reported as the mechanism of disease for *SDHB*-related mitochondrial complex II deficiency. Individuals of reproductive age are at 25% risk of having a child with *SDHB*-related mitochondrial complex II deficiency with each pregnancy when both biological parents have a pathogenic variant in *SDHB*.

Order Summary: The following products were included in the test order for this individual. Please note: tests on hold and those that have been cancelled (including reflex testing steps cancelled due to a positive result in a preceding test) are excluded. For additional information, please contact Ambry Genetics.

- BRCA1/2 seq and del/dup (Product Code 8838)
- CancerNext-Expanded® +RNAinsight® (Product Code 8875-R)

ELECTRONICALLY SIGNED BY

Elizabeth Chao, MD, FACMG, on 10/31/2025 at 09:51:21 am

ASSAY INFORMATION

General methodology: Genomic deoxyribonucleic acid (gDNA) is isolated from the patient's specimen using standardized methodology and quantified. Sequence enrichment of the targeted coding exons and adjacent intronic nucleotides is carried out by a bait-capture methodology using long biotinylated oligonucleotide probes followed by polymerase chain reaction (PCR) and Next-Generation sequencing (NGS). Variants in regions complicated by pseudogene interference, variant calls not satisfying depth of coverage and variant allele frequency quality thresholds, and potentially homozygous variants are verified by Sanger sequencing. Gross deletion/duplication analysis is performed using a customized pipeline using a combination of third-party coverage-based tools and custom methodologies with confirmatory MLPA and/or targeted chromosomal microarray. Mobile element insertions, if detected, are confirmed by PCR and Sanger sequencing and/or gel electrophoresis.

Ribonucleic acid (RNA) is isolated from the patient's specimen using standardized methodology and quantified. RNA is converted to complementary DNA (cDNA) by reverse transcriptase polymerase chain reaction (RT-PCR). Sequence enrichment is carried out by a bait-capture methodology using long biotinylated oligonucleotide probes followed by polymerase chain reaction (PCR) and Next-Generation sequencing (NGS). RNA transcripts are screened and compared to a human reference pool. The presence of RNA transcripts meeting quality thresholds is incorporated as evidence for the assessment and classification of DNA variants. Any region not meeting RNA quality thresholds, including regions with chronically low expression in human peripheral lymphocytes, are excluded from analysis.

Additional methodology:

- ***MSH2***: The inversion of coding exons 1-7 is detected by NGS and confirmed by multiplex ligation-dependent probe amplification (MLPA) or PCR and agarose gel electrophoresis.
- ***PMS2****:* Gross deletions and duplications of exons 11-15 of *PMS2* are reflexed to long-range PCR and gel electrophoresis and/or sequencing to determine if the event occurs within *PMS2* or *PMS2CL*. The most likely deletion/duplication configuration that is consistent with the long-range PCR results is reported; however, rare complex rearrangements in *PMS2* and *PMS2CL* cannot be ruled out.

NCBI reference sequences: *AIP*- NM_003977.2, *ALK*- NM_004304.4, *APC*- NM_000038.5 & NM_001127511.2, *ATM*- NM_000051.3, *ATRIP*- NM_130384.1, *AXIN2*- NM_004655.3, *BAP1*- NM_004656.2, *BARD1*- NM_000465.2, *BMPR1A*- NM_004329.2, *BRCA1*- NM_007294.3, *BRCA2*- NM_000059.3, *BRIP1*- NM_032043.2, *CDC73*- NM_024529.4, *CDH1*- NM_004360.3, *CDK4*- NM_000075.3, *CDKN1B*- NM_004064.3, *CDKN2A*- NM_000077.4 & NM_058195.3, *CEBPA*- NM_004364.3, *CFTR*- NM_000492.3, *CHEK2*- NM_007194.3, *CPA1*- NM_001868.2, *CTNNA1*- NM_001903.2, *CTRC*- NM_007272.2, *DDX41*- NM_016222.2, *DICER1*- NM_177438.2, *EGFR*- NM_005228.3, *EGLN1*- NM_022051.2, *EPCAM*- NM_002354.2, *ETV6*- NM_001987.4, *FH*- NM_000143.3, *FLCN*- NM_144997.5, *GATA2*- NM_032638.4, *GREM1*- NM_013372.6, *HOXB13*- NM_006361.5, *KIF1B*- NM_015074.3, *KIT*- NM_000222.2, *LZTR1*- NM_006767.3, *MAX*- NM_002382.3, *MBD4*- NM_001276270.2, *MEN1*- NM_130799.2, *MET*- NM_001127500.1, *MITF*- NM_000248.3, *MLH1*- NM_000249.3, *MLH3*- NM_001040108.1, *MSH2*- NM_000251.1, *MSH3*- NM_002439.3, *MSH6*- NM_000179.2, *MUTYH*- NM_001128425.1, *NF1*- NM_000267.3, *NF2*- NM_000268.3, *NTHL1*- NM_002528.5, *PALB2*- NM_024675.3, *PALLD*- NM_001166110.1, *PDGFRA*- NM_006206.4, *PHOX2B*- NM_003924.3, *PMS2*- NM_000535.5, *POLD1*- NM_002691.2, *POLE*- NM_006231.2, *POT1*- NM_015450.2, *PRKAR1A*- NM_002734.3, *PRSS1*- NM_002769.4, *PTCH1*- NM_000264.3, *PTEN*- NM_000314.4, *RAD51B* - NM_133510.3, *RAD51C*- NM_058216.1, *RAD51D*- NM_002878.3, *RB1*- NM_000321.2, *RET*- NM_020975.4, *RNF43*- NM_017763.4, *RPS20*- NM_001023.3, *RUNX1*- NM_001754.4, *SDHA*- NM_004168.2, *SDHAF2*- NM_017841.2, *SDHB*- NM_003000.2, *SDHC*- NM_003001.3, *SDHD*- NM_003002.2, *SMAD4*- NM_005359.5, *SMARCA4*- NM_001128849.1, *SMARCB1*- NM_003073.3, *SMARCE1*- NM_003079.4, *SPINK1*- NM_003122.3, *STK11*- NM_000455.4, *SUFU*- NM_016169.3, *TERT* - NM_198253.2, *TMEM127*- NM_017849.3, *TP53*- NM_000546.4, *TSC1*- NM_000368.4, *TSC2*- NM_000548.3, *VHL*- NM_000551.3, and *WT1*- NM_024426.4.

Analytical range: This test detects variants in the coding domains and well into the flanking 5' and 3' ends of the introns and untranslated regions. Unless explicitly stated, sequence and copy number variants in the promoter, non-coding exons, or 3' untranslated regions are not routinely reported.

Analytical range exceptions:

- ***APC***: all promoter 1B gross deletions as well as single nucleotide substitutions within the promoter 1B YY1 binding motif (NM_001127511 c.-196_-186) are analyzed and reported.
- ***EPCAM***: only gross deletions encompassing the 3' end of the gene are reported.
- ***GREM1***: only the status of the 40kb 5'UTR gross duplication is analyzed and reported.
- ***MITF***: only the c.952G>A (p.E318K) variant is reported.
- ***MSH3*** and ***PHOX2B***: the polyalanine repeat regions are excluded from analysis.
- ***NTHL1***: only full-gene gross deletions and duplications are detected.
- **Gross deletion/duplication analysis is not performed for the following genes:** *ATRIP,AXIN2, CFTR, CPA1, CTNNA1, CTRC, DDX41, EGFR, EGLN1, HOXB13, KIT, MBD4, MITF, MLH3, MSH3, PALLD, PDGFRA, POLD1, POLE, PRSS1, RAD51B, RNF43, SPINK1*, and *TERT*.

Reporting: Results reported herein may be of constitutional or somatic origin. This methodology cannot differentiate between these possibilities. In result reports, variants in the following classifications are always reported, and are based on the following definitions and clinical recommendations.

- **Pathogenic Variant**: variants with sufficient evidence to classify as pathogenic (capable of causing disease). Targeted testing of at-risk relatives and appropriate changes in medical management for pathogenic variant carriers recommended. Previously described pathogenic variants, including intronic variants at any position, are always reported when detected.
- **Variant, Likely Pathogenic (VLP)**: variants with strong evidence in favor of pathogenicity. Targeted testing of at-risk relatives and appropriate changes in medical management for VLP carriers typically recommended. Previously described likely pathogenic variants, including intronic VLPs at any position, are always reported when detected.
- **Variant, Uncertain Significance (VUS)**: variants with limited and/or conflicting evidence regarding pathogenicity. Familial testing via the Family Studies Program may be recommended. Medical management to be based on personal/family clinical histories, not VUS carrier status. Note, intronic VUSs are always reported out to 5 base pairs from the splice junction when detected.

Variants of unlikely clinical significance (those with strong/very strong evidence to argue against pathogenicity) are not routinely included in results. These include findings classified as “likely benign” and “benign” variants. Classification and interpretation of variants may change over time with accumulating evidence and scientific advancements. Updated classifications may be reported through reclassification notices; however, clients should re-contact the laboratory or visit ClinVar (https://www.ncbi.nlm.nih.gov/clinvar/) for the most up to date information regarding the current interpretation of results.

RNA transcripts derived from genes with limited gene-disease validity or with an inconsistent mechanism of disease do not routinely contribute to variant interpretation.

All results, including those from prior genetic testing for themselves and/or family members, will be reported as described above.

Gender identity (if provided) is not used in the interpretation of results, and sex assigned at birth is used in the interpretation of results only when necessary. Currently, there are insufficient data to determine specific cancer risk adjustments for transgender, nonbinary, or intersex individuals.

Assay Information Continued on Next Page

ASSAY INFORMATION *(Supplement to Test Results - Continued)*

Resources: The following references are used in variant analysis and classification when applicable for observed genetic alterations.

1. The 1000 Genomes Project Consortium. An integrated map of genetic variation from 1092 human genomes. *Nature*. 2012;491:56-65.
2. ACMG Standards and guidelines for the interpretation of sequence variants. *Genet Med.* 2015 May;17(5):405-23.
3. Ambry Genetics Variant Classification Scheme. http://www.ambrygen.com/variant-classification.
4. Berkeley Drosophila Genome Project [Internet]. Reese MG et al. *J Comp Biol.* 1997;4:311-23. http://www.fruitfly.org/seq_tools/splice.html.
5. Database of Single Nucleotide Polymorphisms (dbSNP) [Internet]. Bethesda (MD): National Center for Biotechnology Information, National Library of Medicine (dbSNP Build ID:135) Available from: www.ncbi.nlm.nih.gov/SNP. Accessed Jan 2012).
6. ESEfinder [Internet]. Smith PJ, et al. (2006) *Hum Mol Genet.* 15(16):2490-2508 and Cartegni L, et al. *Nucleic Acid Research.* 2003;31(13):3568-3571. http://rulai.cshl.edu/cgi-bin/tools/ESE3/esefinder.cgi?process=home.
7. Exome Variant Server, NHLBI Exome Sequencing Project (ESP) [Internet], Seattle WA. Available from: evs.gs.washington.edu/EVS.
8. Grantham R. Amino acid difference formula to help explain protein evolution. *Science.* 1974;185(4151):862-864.
9. HGMD® [Internet]: Stenson PD et al. *Genome Med.* 2009;1(1):13. www.hgmd.cf.ac.uk.
10. Landrum MJ et al. ClinVar: public archive of relationships among sequence variation and human phenotype. *Nucleic Acids Res*. 2014 Jan 1;42(1):D980-5. doi: 10.1093/nar/gkt1113. PubMed PMID: 24234437.
11. Online Mendelian Inheritance in Man, OMIM®. McKusick-Nathans Institute of Genetic Medicine, Johns Hopkins University (Baltimore, MD), Copyright® 1966-2012. World Wide Web URL: http://omim.org.
12. Feng BJ. PERCH: A Unified Framework for Disease Gene Prioritization. *Hum Mutat.* 2017 Mar;38(3):243-251.
13. Exome Aggregation Consortium (ExAC) [Internet], Cambridge, MA. Available from: http://exac.broadinstitute.org.
14. Genome Aggregation Database (gnomAD) [Internet], Cambridge, MA. Available from: http://gnomad.broadinstitute.org.
15. Lek M et al. Analysis of protein-coding genetic variation in 60,706 humans. *Nature*. 2016 Aug 17;536(7616):285-91. PMID: 27535533
16. Mu W et al. *J Mol Diagn*. 2016 Oct 4. PubMed PMID: 27720647
17. Karczewski KJ et al. *Nature*. 2020 May;581(7809):434-443. PMID: 32461654
18. Splicing Prediction: Jaganathan K et al. *Cell.* 2019 Jan 24; 176(3):535-548.e24. PMID: 30661751

Disclaimer: This test was developed, and its performance characteristics were determined by Ambry Genetics Corporation. It has not been cleared or approved by the US Food and Drug Administration (FDA). The FDA does not require this test to go through premarket FDA review. It should not be regarded as investigational or for research. This test should be interpreted in context with other clinical findings. This report does not represent medical advice. Any questions, suggestions, or concerns regarding interpretation of results should be forwarded to a genetic counselor, medical geneticist, or physician skilled in interpretation of the relevant medical literature. This laboratory is certified under the Clinical Laboratory Improvement Amendments (CLIA) as qualified to perform high complexity clinical laboratory testing. This test analyzes the following types of mutations: nucleotide substitutions, small deletions (up to 25 bp), small insertions (up to 10 bp), small indels, and gross deletions/duplications. Unless otherwise noted in the methodology section above, this test is not intended to analyze the following types of alterations: gross rearrangements, deep intronic variations, mobile element insertions, and other unknown abnormalities. The pattern of mutation types varies by gene, and this test detects a high but variable percentage of known and unknown mutations of the classes stated. A negative result from the analysis cannot rule out the possibility that the tested individual carries a rare unexamined mutation or mutation in the undetectable group. This test is designed and validated to be capable of detecting ~99.9% of described mutations in the genes represented on the test, listed above (analytical sensitivity). The clinical sensitivity of this test may vary widely according to the specific clinical and family history. Mutations in other genes or the regions not analyzed by this test can also give rise to similar clinical conditions. Although molecular tests are highly accurate, rare diagnostic errors may occur. Possible diagnostic errors include sample mix-up, erroneous paternity identification, technical errors, clerical errors, and genotyping errors. Genotyping errors can result from trace contamination of PCR reactions, from maternal cell contamination in fetal samples, from rare genetic variants that interfere with analysis, germline or somatic mosaicism, presence of pseudogenes, technical difficulties in regions with high GC content or homopolymer tracts, active hematologic disease, a history of allogeneic bone marrow or peripheral stem cell transplant, or from other sources. Rare variants present in the human genome reference sequence (GRCh37.p5/hg19) or rare misalignment due to presence of pseudogenes can lead to misinterpretation of patient sequence data.

TP53 NM_000546 c.460G>A p.G154S

VARIANT DETAILS:

The **p.G154S** variant (also known as c.460G>A), located in coding exon 4 of the *TP53* gene, results from a G to A substitution at nucleotide position 460. The glycine at codon 154 is replaced by serine, an amino acid with similar properties. This alteration has been reported in a patient diagnosed with adrenocortical carcinoma at one year of age (Butz H et al. *Sci Rep*, 2023 Aug;13:14259). This alteration has also been reported in a patient with breast cancer at age 38 and in a patient with renal cancer at age 40 (Yang XR et al. *Breast Cancer Res.Treat.* 2017 Oct;165(3):687-697; Bittar CM et al. *Fam Cancer*, 2019 10;18:451-456). This variant is in the DNA binding domain of the TP53 protein and is reported to have partial loss of transactivation capacity and predicted to affect several p53 isoforms (Kato S et al. *Proc Natl Acad Sci USA*. 2003 Jul 8;100(14):8424-9). Studies conducted in human cell lines indicate this alteration remains proficient at growth suppression (Kotler E et al. *Mol.Cell,* 2018 Jul;71:178-190.e8; Giacomelli AO et al. *Nat. Genet.,* 2018 Oct;50:1381-1387). This amino acid position is well conserved in available vertebrate species. In addition, this alteration is predicted to be inconclusive by *in silico* analyses. Since supporting evidence is limited at this time, the clinical significance of this alteration remains unclear.

FAMILY STUDIES PROGRAM:

Ambry Genetics offers complimentary genetic studies for variants of unknown significance (VUSs) meeting specific criteria in appropriate family members. Review of clinical information is required. Additional information, application instructions and required forms, and patient education materials are available at http://ambrygen.com/family-studies-program. For additional information, please email us at GeneticCounselor@ambrygen.com or call 949-900-5500 and ask to speak with a genetic counselor.

Please note that the classification of variants may change over time as additional information becomes available. Alerts are disseminated via fax and/or AmbryPort email to clinicians upon clinically relevant variant reclassifications. If no updates are received, clinicians are encouraged to contact the laboratory at 949-900-5500 once a year to review the status of previously reported variants.

GENE INFORMATION:

The *TP53* gene (NM_000546.4) is located on chromosome 17p13.1, encodes the cellular tumor antigen p53 protein, and contains ten coding exons. Germline pathogenic variants in this gene have been associated with a spectrum of tumor predisposition syndromes including classic Li-Fraumeni syndrome (LFS) and attenuated LFS, which are inherited in an autosomal dominant fashion, and *TP53*-related bone marrow failure, which is an autosomal dominant condition that generally occurs *de novo*. Classic LFS is defined by three criteria, all of which must be satisfied: an individual diagnosed less than 45 years with a sarcoma, a first-degree relative diagnosed less than 45 years with any cancer, and an additional first- or second-degree relative with cancer diagnosed less than 45 years (or a sarcoma at any age). Germline pathogenic variants in *TP53* confer significantly increased cumulative lifetime risks for four core cancers, including pre-menopausal female breast cancer (50-85%), soft tissue sarcoma (6-22%), brain cancer (5-19%), and osteosarcoma (5-11%). Risks are also elevated for colorectal, gastric, and prostate cancers as well as leukemia and melanoma. Childhood cancers (usually diagnosed prior to 15 years) strongly associated with LFS include adrenocortical carcinoma (ACC), medulloblastoma, rhabdomyosarcoma, hypodiploid acute lymphoblastic leukemia, and choroid plexus carcinoma. Lifetime risks for any cancer have been estimated at approximately 92% in males and 98% in females, and there is a significantly elevated risk of multiple primary cancers in carriers of germline *TP53* pathogenic variants. Attenuated LFS is characterized by an increased risk for LFS-related tumors, but these individuals do not meet classic LFS criteria; risks for first and subsequent cancers may be less than in individuals with classic LFS. It has been suggested that the median age of cancer diagnosis in individuals with attenuated LFS is approximately 44 years, compared to the median age of diagnosis of 26 years in individuals with classic LFS (Fortuno C, et al. (2024) *JCO Precis Oncol* 8(0):e2300453.; Guha T and Malkin D. *Cold Spring Harb Perspect Med.* 2017 Apr 3;7(4):a026187; Mai PL et al. *Cancer.* 2016 Dec 1:122(23):3673-3681; Amadou A et al. *Curr Opin Oncol.* 2018 Jan;30(1):23-29; Kratz C et al. *JAMAOncol.* 2021 Dec 1;7(12):1800-1805; Wu CC et al.?*Cancer Res.* 2006 Aug 15;66(16):8287-92; Rana HQ et al. *J Natl Cancer Inst.* 2018 Aug 1;110(8):863-870; Ambry Internal Data). Penetrance in individuals and families even with the same pathogenic *TP53* variant is variable; therefore, cancer risks will differ based on individual and family history. In addition, *TP53* pathogenic variants identified in peripheral blood or saliva are not always representative of an individual's germline and may represent an incidental finding on genetic testing. Acquired somatic *TP53* variants can occur as a result of aberrant clonal expansion (ACE), most often due to a phenomenon known as clonal hematopoiesis (CHiP), which occurs more frequently in healthy older (age 65 and older) populations. Somatic *TP53* variants may also be detected in blood or saliva as a result of chemotherapy or radiation treatments. Individuals with somatic *TP53* variants do not have LFS and are not at risk for LFS-related cancers; however, this is some evidence that these individuals may be at risk for cardiovascular disease or hematologic malignancies (Weitzel J et al. *Genet Med.* 2018 Aug;20(8):809-816; Schwartz A et al. *JCO Precis Oncol.* 2021 Nov;5:1677-1686; Chao E et al. *Genet Med* 2021 Jul;23(7):1179-1184). Up to 20% of germline pathogenic variants in *TP53* occur *de novo* (Renaux-Petel M et al. *J Med Genet.* 2018 Mar;55(3):173-180). Loss of function has been reported as the mechanism of disease for LFS. *TP53*-related bone marrow failure is characterized by red cell aplasia, anemia, immune deficiency, global developmental delay, and microcephaly. Other features seen in a minority of patients include reticular skin pigmentation and seizures (Toki T, et al. (2018) *Am J Hum Genet* 103(3):440-447; Fedorova D, et al. (2022) *Pediatr Blood Cancer* 69(4):e29558; Kumar RD, et al. (2022) *Am J Med Genet* A 188(7):2204-2208). *De novo* C-terminal premature truncation variants not expected to undergo nonsense mediated decay have been reported as disease-causing for *TP53*-related bone marrow failure.

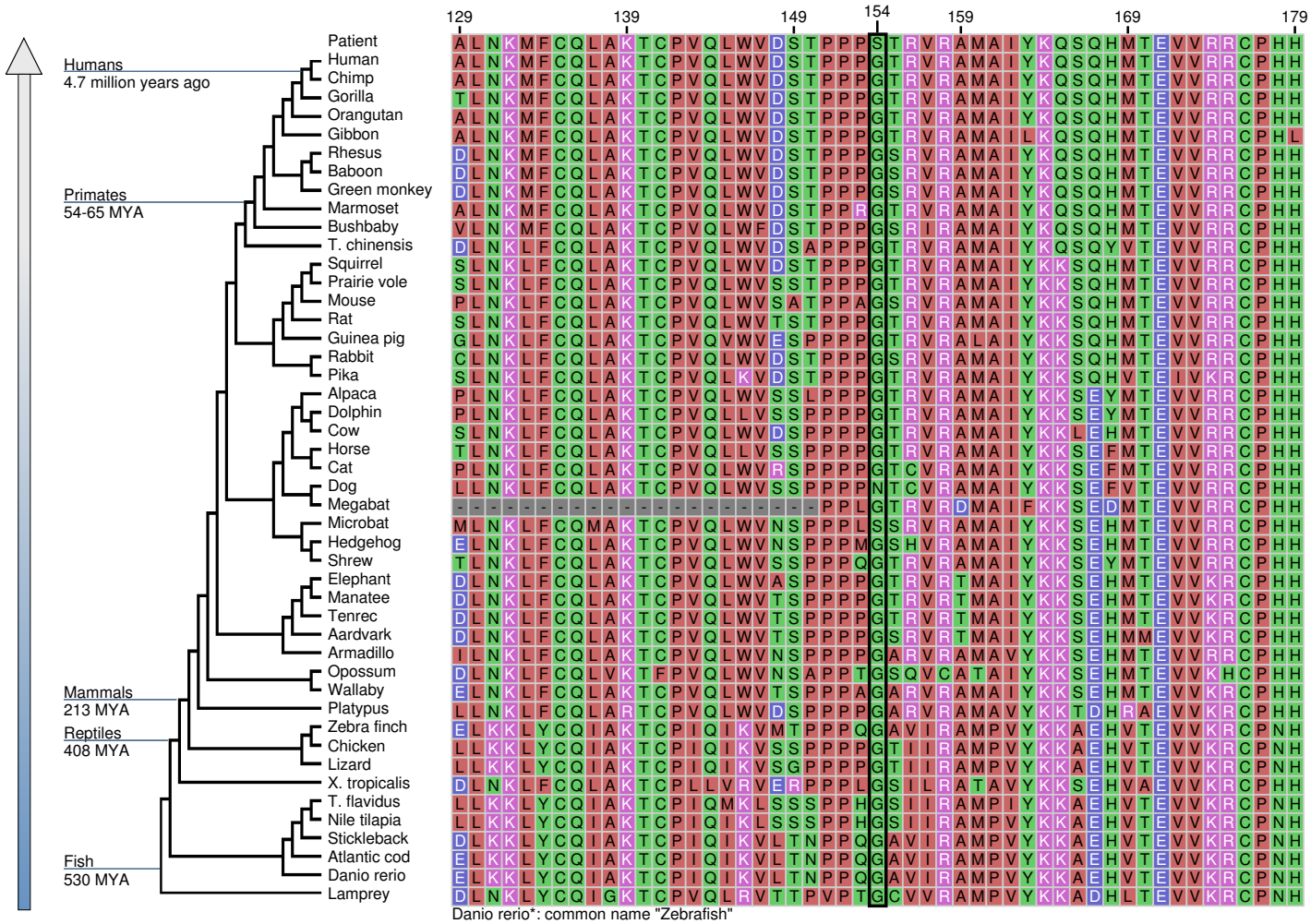
ADDITIONAL SUPPORTING INFORMATION:

Co-Segregation	Co-segregation data for this variant is currently unavailable.
Co-occurrence	No significant co-occurrence data is currently available at our laboratory.
Frequency	Internal Frequency: <0.01% (15/488000) total alleles studied.
	gnomAD: <0.01% (6/282632) total alleles studied, <0.01% (6/129022) European (non-Finnish) alleles.
Grantham Score	56 (similar amino acid substitution)
in silico	Inconclusive

TP53 NM_000546 c.460G>A p.G154S

Evolutionary conservation diagram: Amino Acid Alignment

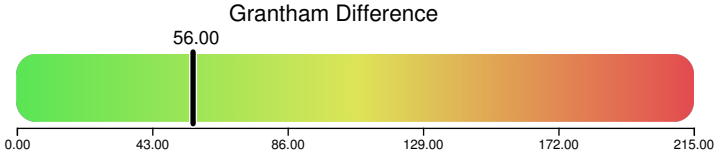
This amino acid position is well conserved in available vertebrate species.



Danio rerio*: common name "Zebrafish"

Amino Acid Change:

Trait	Gly (G)	Ser (S)
Amino Acid Name	Glycine	Serine
Polarity/Charge	non-polar	polar
pH	neutral	neutral
Residue Weight	57	87
Hydrophobicity Score	-0.4	-0.8
Hydrophilicity Score	0	0.3
Secondary Structure Propensity	strong α breaker / β breaker	α indifferent / β breaker



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Clinician Management Resource for Hereditary Pheochromocytoma/Paraganglioma Syndrome

This overview of clinical management guidelines is based on this patient's positive test result. Unless otherwise stated, medical management guidelines used here are limited to those issued by the National Comprehensive Cancer Network (NCCN) in the U.S. Please consult the referenced guideline for complete details and further information.

Clinical correlation with the patient's past medical history, treatments, surgeries, and family history may lead to changes in clinical management decisions; therefore, other management recommendations may be considered. Genetic testing results and medical society guidelines help inform medical management decision but do not constitute formal recommendations. Discussions of medical management decisions and individualized treatment plans should be made in consultation between each patient and his or her healthcare provider and may change.

SCREENING CONSIDERATIONS[1, ^, *]	AGE TO START	FREQUENCY
Pheochromocytoma/Paraganglioma (PCC/PGL)-specific screening recommendations for patients with confirmed germline hereditary PCC/PGL syndrome		
Blood pressure monitoring.	6-10 years old for patients with pathogenic or likely pathogenic *SDHB* variants 10-15 years for patients with all other forms of hereditary PCC/PGL	At all medical visits
Measurement of plasma free metanephrines or 24-hour urine for fractionated metanephrines.	6-10 years old for patients with pathogenic or likely pathogenic *SDHB* variants 10-15 years for patients with all other forms of hereditary PCC/PGL	Annually
Cross-sectional imaging of skull base to pelvis. Whole body MRI or other non-radiation-containing imaging procedures. If whole body MRI not available, may consider abdominal MRI, skull base and neck MRI, and chest CT.**	6-10 years old for patients with pathogenic or likely pathogenic *SDHB* variants 10-15 years for patients with all other forms of hereditary PCC/PGL	Every 2–3 years

^ Patients with pathogenic *SDHD*, *SDHAF2*, or *MAX* variants are most at risk if the pathogenic variant was paternally inherited. Recommend following the above recommendations if the parent of origin is unknown. Consider screening for patients with maternally inherited variants as case reports of tumor occurrence exist.

* If asymptomatic and without a prior history of elevation, annual follow-up and testing can be omitted or done with imaging every 2-3 years. Since SDH genes have variability in their tumor penetrance and risk for malignancy, consideration can be given to modified screening intervals, especially for less penetrant genes such as *SDHA*.

** Available data suggest that patients with pathogenic *SDHAF2* variants are primarily at risk for head and neck tumors and patients with pathogenic *MAX* variants are primarily at risk for adrenal tumors. Therefore, consideration can be given to more targeted imaging in these cohorts.

1. Referenced with permission from the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines) for Neuroendocrine and Adrenal Tumors v1.2025. National Comprehensive Cancer Network, Inc. 2025. All rights reserved. Accessed March 31, 2025. To view the most recent and complete version of the guideline, go to NCCN.org. NCCN makes no warranties of any kind whatsoever regarding their content, use or application and disclaims any responsibility for their application or use in any way.

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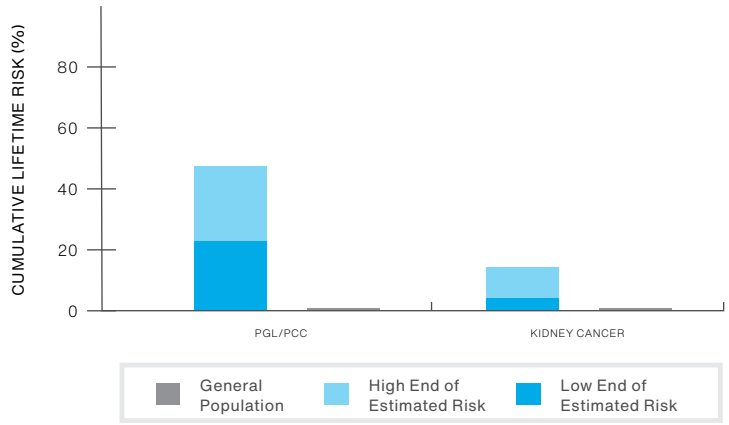
Understanding Your Positive *SDHB* Genetic Test Result

INFORMATION FOR PATIENTS WITH A PATHOGENIC OR LIKELY PATHOGENIC VARIANT

5 Things to Know

1	Result	Your testing shows that you have a pathogenic or likely pathogenic variant in the *SDHB* gene.
2	Non-cancerous tumor and cancer risks	You have an increased chance to develop pheochromocytomas (PCCs)/paragangliomas (PGLs), gastrointestinal stromal tumors (GISTs), and kidney cancer.
3	What you can do	Risk management decisions are very personal. There are options to detect cancer early or lower the risk to develop cancer. It is important to discuss these options with your healthcare provider and decide on a plan that works for you.
4	Other medical concerns	Individuals with a pathogenic or likely pathogenic *SDHB* variant may have an increased risk to have a child with mitochondrial complex II deficiency, but only if their partner also carries a pathogenic or likely pathogenic variant in the *SDHB* gene. Mitochondrial complex II deficiency is a rare, highly variable autosomal recessive condition that can can affect many different parts of the body, including the brain, heart, and muscles.
5	Family	Family members may also be at risk – they can be tested for the pathogenic or likely pathogenic *SDHB* variant that was found in you. It is recommended that you share this information with your family members so they can learn more and discuss with their healthcare providers.

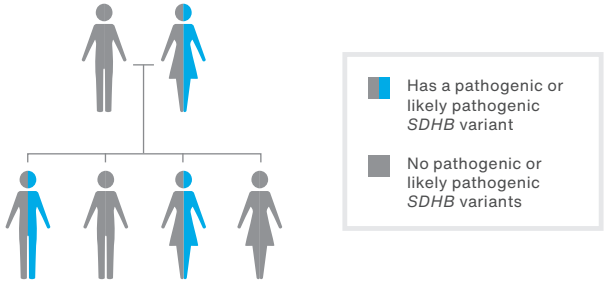
SDHB Lifetime Cancer Risks*



* Because risk estimates vary in different studies, only approximate risks are given. Cancer risks will differ based on individual and family history.

SDHB in the Family

There is a 50/50 random chance to pass on the pathogenic or likely pathogenic *SDHB* variant to each of your children.



RESOURCES

- Pheo Para Alliance pheopara.org
- National Society of Genetic Counselors nsgc.org
- Canadian Society of Genetic Counsellors cagc-accg.ca

Please discuss this information with your healthcare provider. The cancer genetics field is continuously evolving, so updates related to your *SDHB* result, medical recommendations, and/or potential treatments may be available over time. This information is not meant to replace a discussion with a healthcare provider, and should not be considered or interpreted as medical advice.

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PROMPT

Prospective Registry Of MultiPlex Testing

Opportunity to Enroll in Hereditary Cancer Research

Genetic testing can help individuals and families by giving them a clearer idea of their cancer risks. Genetic tests (called multi-gene or multiplex panels) look for changes in several different genes, all in a single test. While all of the genes on these panels have been tied to an increased risk of cancer, we understand the risks associated with some of the genes better than we understand others. One way to help improve our understanding is to enroll people with pathogenic mutations or variants of unknown significance in registries. Registries typically follow people over many years to learn more about these alterations and how they impact their health.

How can I find a research registry?

There are several hereditary cancer research registries that are studying individuals who have had multiplex panel testing. One registry that is open to individuals nationwide is PROMPT (or **P**rospective **R**egistry **O**f **M**ulti**P**lex **T**esting). PROMPT is an online registry for patients and families who have had multiplex testing and have been found to have a genetic variation which may be linked to an increased risk of cancer. PROMPT is a joint effort involving several academic medical centers and commercial laboratories, working together to learn more about the genes that are studied on multiplex panels. PROMPT will allow researchers to better understand the cancer risks associated with changes in these genes and thus provide a better understanding of the best way to take care of individuals who have such changes.

What is involved in participation?

Participation in the study simply involves completing online surveys. Additionally, the PROMPT team may reach out to you to talk about ways that you can get more involved with the research effort. Your participation will help researchers learn more and improve the ability of this genetic testing to help people.

How do I enroll?

You can learn more about or register for PROMPT by going to www.promptstudy.info or by scanning the QR code below.

Thank you again for considering taking part in PROMPT!

If you would like to read more about multiplex panels, including details about specific genes, please visit our informational website at www.promptstudy.info.

Opportunity to connect and help prevent cancer in your family

Did you recently have genetic testing for a cancer gene variant (or mutation) known to be in your family? Questions such as “Where did this variant come from?” or “What can I do to help others in my family?” are common. ConnectMyVariant can help!

ConnectMyVariant provides resources for people who want help talking with relatives about cancer risk or finding new relatives who might be at risk to help them get genetic testing and prevent cancer.

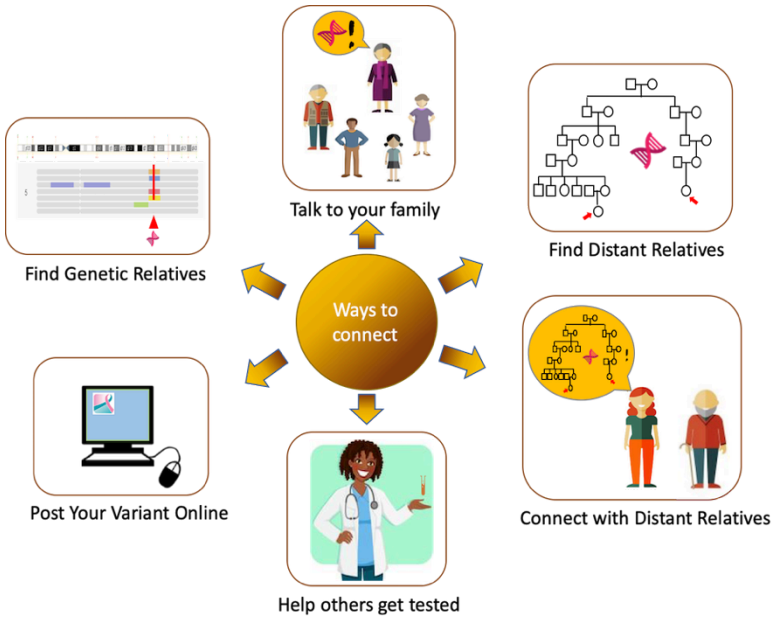
The ConnectMyVariant initiative seeks to help people like you:

- Talk to their relatives
- Share important genetic information
- Expand family trees to identify and connect with distant at-risk relatives
- Guide at-risk relatives to cancer prevention.

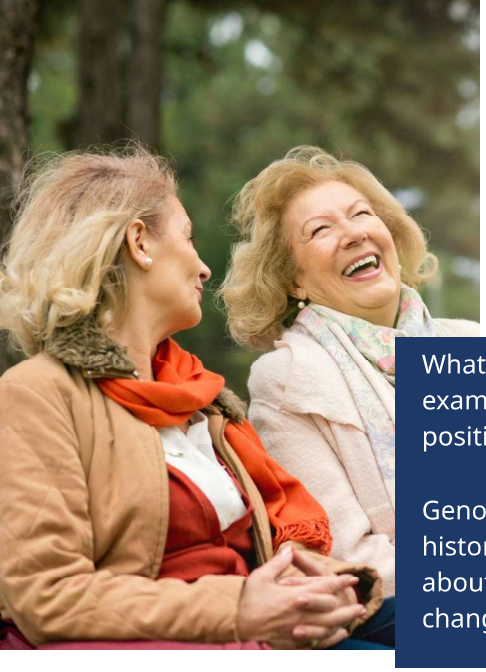
“Prevention Through Connection”

People with the same genetic variant may be distantly related through a long-ago ancestor. This means that your family’s variant may be a key to understanding your family’s past. It is also a key that you can use to help both close and distant family members prevent cancer before it happens.

You may have received genetic testing because someone cared enough to warn you about your risk. Now you can find and warn other at-risk relatives. Reaching out and speaking to other at-risk relatives to help them get genetic testing may help prevent cancer and save lives. These are the goals of ConnectMyVariant.



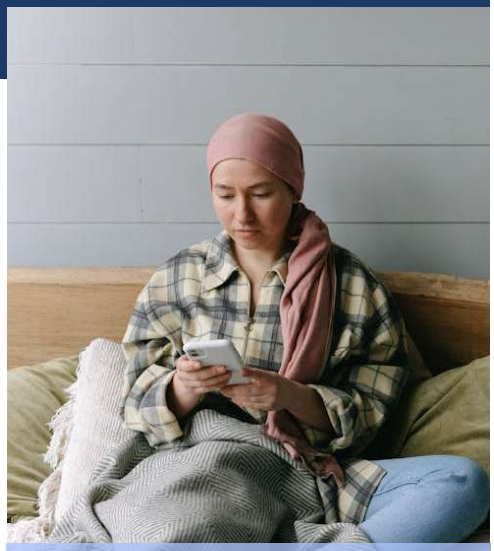
You can learn more and sign up at http://connectmyvariant.org/
Questions? info@connectmyvariant.org



GenomeConnect
The ClinGen Patient Portal

What we know about genetic changes (also called variants) can change over time. For example, a result that is classified as uncertain (called a VUS) might later be changed to a positive or negative result.

GenomeConnect is a registry that helps you share your genetic test results and health history. Sharing your genetic changes and health history helps scientists learn more about genes and health and can help identify changes in variant classification over time.



Who can participate in GenomeConnect?

Anyone who has had genetic testing can join GenomeConnect. It doesn't matter what your results were—positive, negative, or uncertain (VUS). People with results in any gene can take part. GenomeConnect is available in English and Spanish.

Why join GenomeConnect?

Sometimes, when a lab finds a genetic change, they don't know what it means for health. These are called "uncertain" or "VUS" results. Even when a change is well understood, more information can help scientists learn more about the condition and how to treat it.

By joining GenomeConnect, you can securely share your genetic test results and health history with databases like ClinVar and with experts from ClinGen. ClinGen has a focus in cancer including through expert panels working to better understand genetic changes that increase cancer risk.

The more people who share their information, the more scientists can learn about how genes and genetic variants affect health and, ultimately, how best to care for patients.

By participating in GenomeConnect, you can:

- Share your information securely and safely.
- Help **increase our understanding of genetics and health** that might help you and other families.
- Have the option to **receive genetic result updates** from GenomeConnect. This can be particularly helpful for individuals with uncertain (VUS) results.
- Learn about other relevant **research opportunities.**
- **Connect with other participants** with similar results.

What are the steps to participate in GenomeConnect?

Participation is simple

- Sign up and consent www.genomeconnect.org
- Share a copy your genetic test results (no new sample needed)
- Fill out surveys to share your health history
- Update your information once a year

"It's just reassuring, really comforting to get updates and that I don't have to worry about it and just to know that the genetic industry or the genetic world, there is someone that we can rely on."

- GenomeConnect Participant discussing the option to receive genetic result updates

Learn More and Register:
www.genomeconnect.org

Questions:
info@genomeconnect.org

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ClinGen
Clinical Genome Resource

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